How to plot a linear graph

The graph of a linear equation can be drawn by working out several different sets of values for x and y and then plotting these values on a pair of axes. The x values are measured along the x axis, and the y values along the y axis.

▷ **The equation**
This shows that each of the y values for this equation will be double the size of each of the x values.

this means 2 multiplied by x

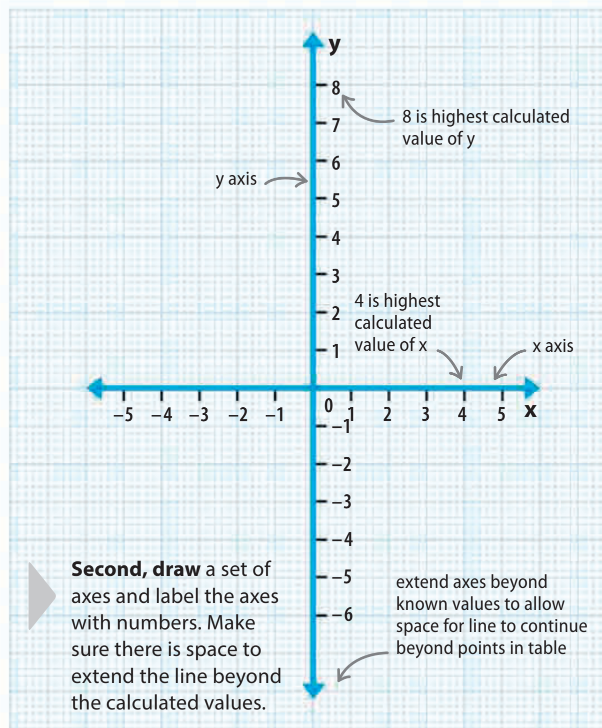
$$y = 2x$$

first, choose some possible values of x

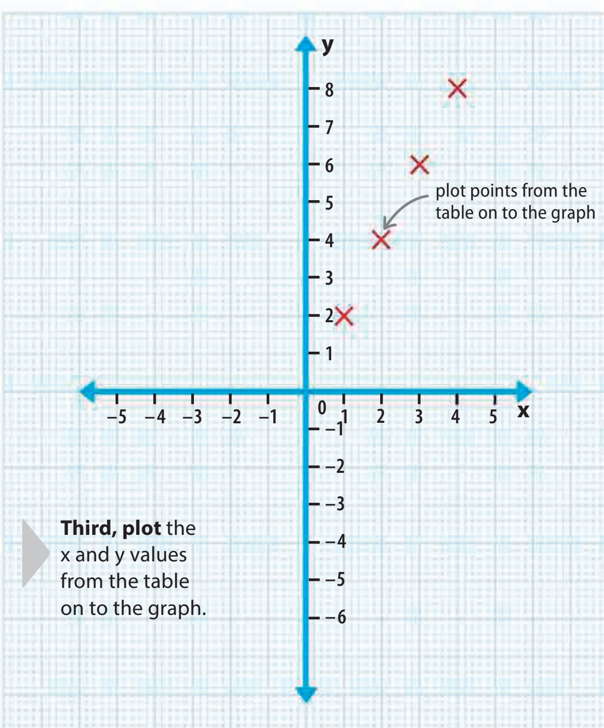
x	y = 2x
1	2
2	4
3	6
4	8

then find corresponding values of y by doubling each x value

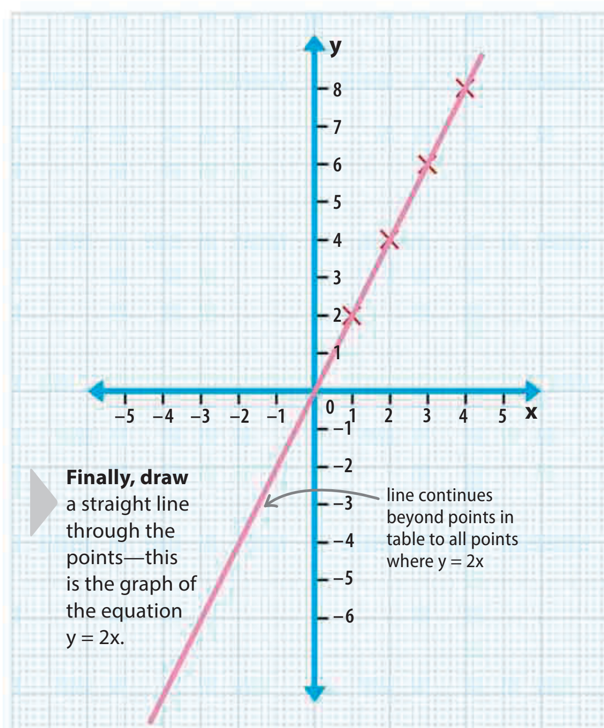
First, choose some possible values of x—numbers below 10 are easiest to work with. Find the corresponding values of y using a table. Put the x values in the first column, then multiply each number by 2 to find the corresponding values for y.



Second, draw a set of axes and label the axes with numbers. Make sure there is space to extend the line beyond the calculated values.



Third, plot the x and y values from the table on to the graph.



Finally, draw a straight line through the points—this is the graph of the equation y = 2x.